# Lab 7: Fuzzing for Input Validation Bugs

Application: atm_withdrawal.py | Tests: test_atm_withdrawal.py (Hypothesis)

### *Screenshot 1: Buggy application output*

Running python atm_withdrawal.py on the original code with input 'abc' crashes with a ValueError.

```
PS C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7> python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): abc
Traceback (most recent call last):
  File "C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\atm_withdrawal.py", line 29, in <module>
    main()
  File "C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\atm_withdrawal.py", line 15, in main
    result = notes_needed(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\atm_withdrawal.py", line 2, in notes_needed
    return int(amount.replace(',', '')) // 100
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: invalid literal for int() with base 10: 'abc'
```

### *Screenshot 2: Hypothesis tests failing on the buggy code*

pytest test_atm_withdrawal.py: 3 failed, 3 passed. Hypothesis found crashing inputs such as '' and None.

```
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\test_atm_withdrawal.py", line 26, in test_average_withdrawal_no_crash
    |     assert False, f"average_withdrawal crashed with: {e!r}"
    | AssertionError: average_withdrawal crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_average_withdrawal_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\test_atm_withdrawal.py:25
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\test_atm_withdrawal.py", line 24, in test_average_withdrawal_no_crash
    |     average_withdrawal(s)
    |   File "C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\atm_withdrawal.py", line 9, in average_withdrawal
    |     amounts = [int(a) for a in amounts_csv.split(',')]
    |                               ^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\test_atm_withdrawal.py", line 26, in test_average_withdrawal_no_crash
    |     assert False, f"average_withdrawal crashed with: {e!r}"
    | AssertionError: average_withdrawal crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_average_withdrawal_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7\test_atm_withdrawal.py:25
    +------------------------------------
=========================================================== short test summary info ============================================================
FAILED test_atm_withdrawal.py::test_notes_needed_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_atm_withdrawal.py::test_mask_card_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_atm_withdrawal.py::test_average_withdrawal_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================================ 3 failed, 3 passed in 8.47s ============================================================
```

### *Screenshot 3: Fixed application output*

After hardening, the same invalid inputs print 'Invalid input' messages instead of crashing.

```
PS C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7> python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): abc
Number of 100 Notes: Invalid input

==== Running mask_card ====
Enter a card number (e.g. 1234 5678 9012 3456): 1234 5678
Masked Card: Invalid card number

==== Running average_withdrawal ====
Enter withdrawals as CSV (e.g. 500,1000,2000):
Average Withdrawal: Invalid input
PS C:\Users\Aditya Shetty\OneDrive\Desktop\lab7_work\Lab 7>
```

## *Screenshot 4: Tests passing on the fixed code*

pytest test_atm_withdrawal.py: all 6 tests passed.

